

100 Days of Azure DevOps

Day 15 Handout — Merge, rebase, squash & history hygiene

Learning in public · Personal lab only · Educational content · Not a sales pitch

Architecture A — history operations compared

Operation	Effect on history	Best used for
Merge	Keeps topology + merge commit	Preserve exact integration story
Rebase	Rewrites SHAs; linear history	Update feature branch onto latest main
Squash	Many commits → one commit	Ship one coherent change via PR
Cherry-pick	Copies a single commit SHA	Hotfix onto another branch line

Architecture B — history hygiene for CI/CD

Rule	Do this	Never do this
Shared branches	Merge into main via PR	Rebase / rewrite published main
Feature branches	Rebase onto origin/main before PR	Leave stale base with surprise conflicts
Force push	git push --force-with-lease	Blind git push --force
Conflicts	Resolve + test, then continue	Abort and force-merge without reading
PR to main	Squash or atomic commits	Dump 30 WIP commits onto main

One-liner to remember

Rebase rewrites history · Merge preserves the plot · Choose based on whether teammates already depend on that timeline

Step-by-step lab (20–30 min)

In your local clone of **azure-100-labs**:

- 1 Create branch: **git switch -c feature/day15-rebase-lab**
- 2 Make 2–3 small commits (notes / docs)
- 3 Fetch and rebase: **git fetch origin && git rebase origin/main**
- 4 If conflicts: fix → **git add** → **git rebase --continue**
- 5 Push safely: **git push --force-with-lease**
- 6 Open PR; prefer squash merge for WIP cleanup
- 7 Document choice in **docs/merge-strategy.md**

Core commands

```
git fetch origin
git rebase origin/main
git rebase --abort # if needed
git push --force-with-lease
git cherry-pick <sha>
```

Done checklist

- I can explain merge vs rebase vs squash
- Feature branch rebased onto origin/main
- Used --force-with-lease (not blind --force)
- Merge strategy notes documented

Tomorrow — Day 16

Git hooks & pre-commit — local gates before the server gates.

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